

Power BI – DAX Date Functions

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Date operation in Microsoft Excel is comparable to those in Data Analysis Expressions (DAX). However, the DateTime data types utilized by Microsoft SQL Server form the foundation of DAX functions. Microsoft Power BI uses the scripting language Data Analysis Expressions (DAX) to create calculated columns, measurements, and custom tables. It offers a number of useful functions that help create some very cool formulas and expressions that can be applied to a number of data analysis issues.

DAX Date Functions

You can develop reliable and high-performance data models in Power BI by having a fundamental understanding of when and how to use the DAX functions. Your efforts to evaluate data will be more effective the more familiar you are with DAX functions. This is due to your ability to quickly access and analyze enormous volumes of data from several tables.

We need to load the data that contains the Date format field. Here we are taking the records of 50 products sold by a library supply company. **Dataset Used: Sheet1**

File	Home	Help	Table tools
Name Sheet1	Mark as date table	Manage relationships	New measure Quick measure New column New table
Structure	Calendars	Relationships	Calculations

Image	Order No.	Order Date (dd-mm-yyyy)
1	521	19-01-1997
2	521	19-01-1997
3	521	19-01-1997
4	521	19-01-1997
5	-	19-01-1992
6	-	19-01-1999
7	-	19-01-1997
8	RDG/Fur/057	19-01-1999
9	-	16-10-2007
10	-	19-01-1992
11	-	19-01-1991
12	-	18-11-2013
13	-	18-11-2013
14	-	18-11-2013
15	4301/HC-Store-Lab	25-02-1989
16	4301/HC-Store-Lab	25-02-1989
17	4301/HC-Store-Lab	25-02-1989
18	H.C/Pur./Order/417010	11/1/1989
19	H.C/Pur./Order/417010	11.01.1989
20	H.C/Pur./Order/417010	11.01.1989
21	H.C/Pur./Order/417010	11.01.1989
22	H.C/Pur./Order/417010	11.01.1989
23	H.C/Pur./Order/417010	11.01.1989

Table: Sheet1 (50 rows)

Then go to New Table Tab, which would contain the output column of the following queries.

File	Home	Help	Table tools
Name Sheet1	Mark as date table	Manage relationships	New measure Quick measure New column New table
Structure	Calendars	Relationships	

Write a DAX expression to create a new table.

DAX Calendar

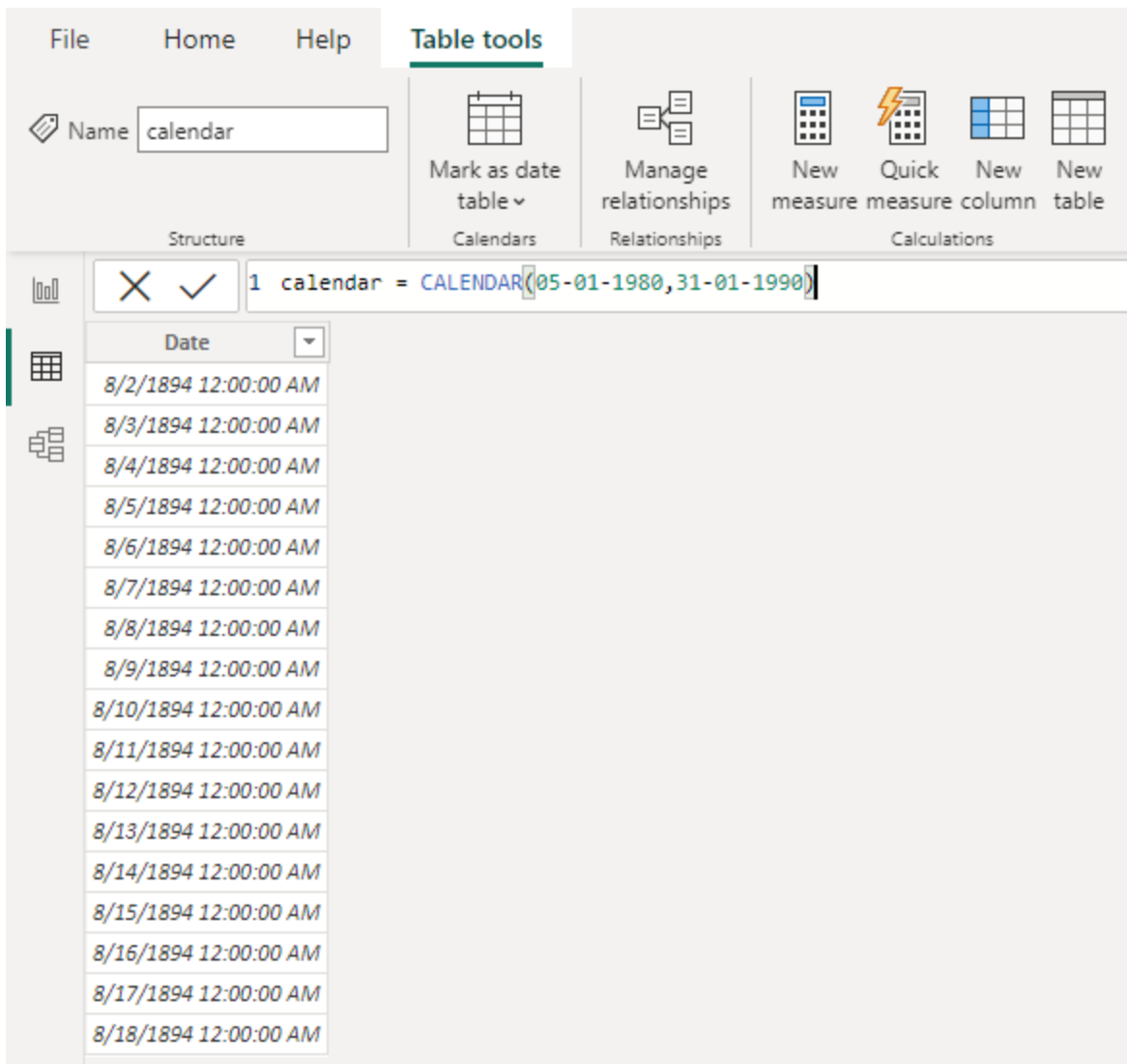
Generates a table with a single “Date” column that is made up of a continuous range of dates. The range of dates is inclusive of those two dates and extends from the specified start date to the specified end date.

Syntax:

```
| CALENDAR(<start_date>, <end_date>)
```

Example:

```
| calendar = CALENDAR(05-01-1980,31-01-1990)
```



DAX Date

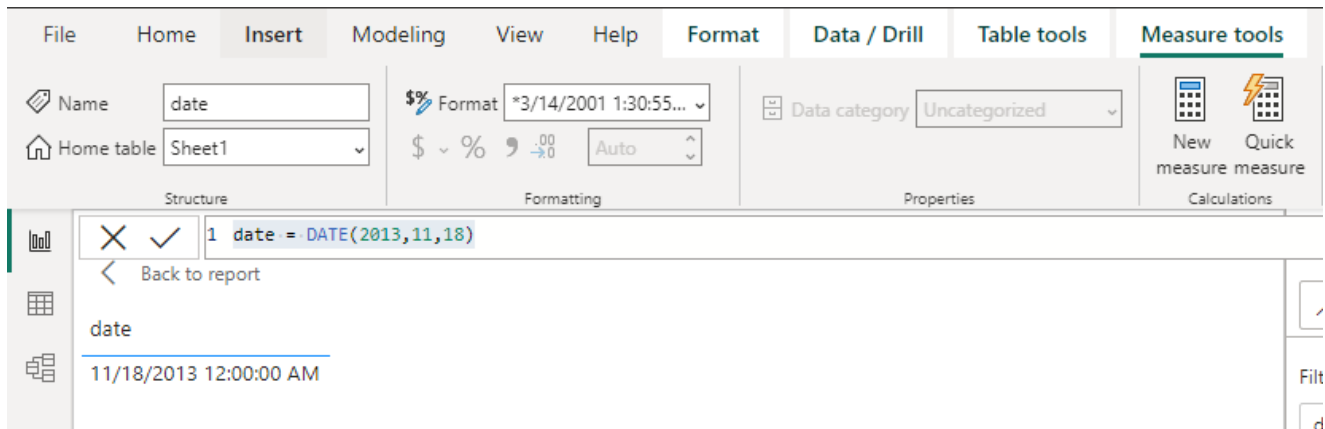
The supplied date is returned in DateTime format by the DATE function. It creates the corresponding date using the integers provided as parameters. When the year, month, and day are determined by formulas, the DATE function is most helpful. The underlying data, for instance, can include dates in a format that isn't accepted as a date, such as YYYYMMDD.

Syntax:

| DATE(<year>, <month>, <day>)

Example:

| date = DATE(2013,11,18)



DAX Datevalue

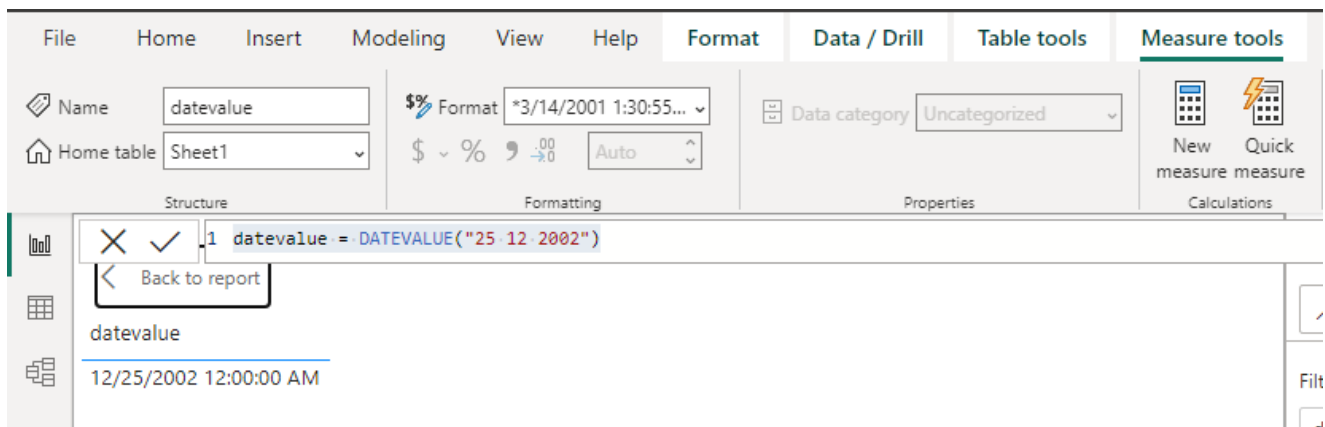
Translates a textual date into a date in DateTime format. The client computer's locale and date/time settings are used by the DATEVALUE function to interpret the text value while executing the conversion.

Syntax:

| DATEVALUE(date_text)

Example:

| datevalue = DATEVALUE("25 12 2002")



DAX Datediff

Returns the number of boundaries between the time frame to compare dates with. Any of the following values may be the value of the Interval:

- Second,
- Minute,
- Hour,
- Day,
- Week,
- Month,
- Quarter,
- Year

Syntax:

| DATEDIFF(<Date1>, <Date2>, <Interval>)

Example:

| datediff = DATEDIFF(DATE (2001, 07, 01),DATE (2002, 12, 31),YEAR)

The screenshot shows the Power BI Desktop interface with the 'Measure tools' tab selected. The 'Name' field is set to 'datediff', the 'Home table' is 'Sheet1', and the 'Format' is 'Whole number'. The 'Data category' is 'Uncategorized'. The measure formula bar displays: `1 datediff = DATEDIFF(DATE ( 2001, 07, 01 ),DATE ( 2002, 12, 31 ),YEAR)`. The 'Structure' pane on the left shows a table with one row and one column labeled 'datediff' with the value '1'. The 'Calculations' pane on the right shows 'New measure' and 'Quick measure' buttons.

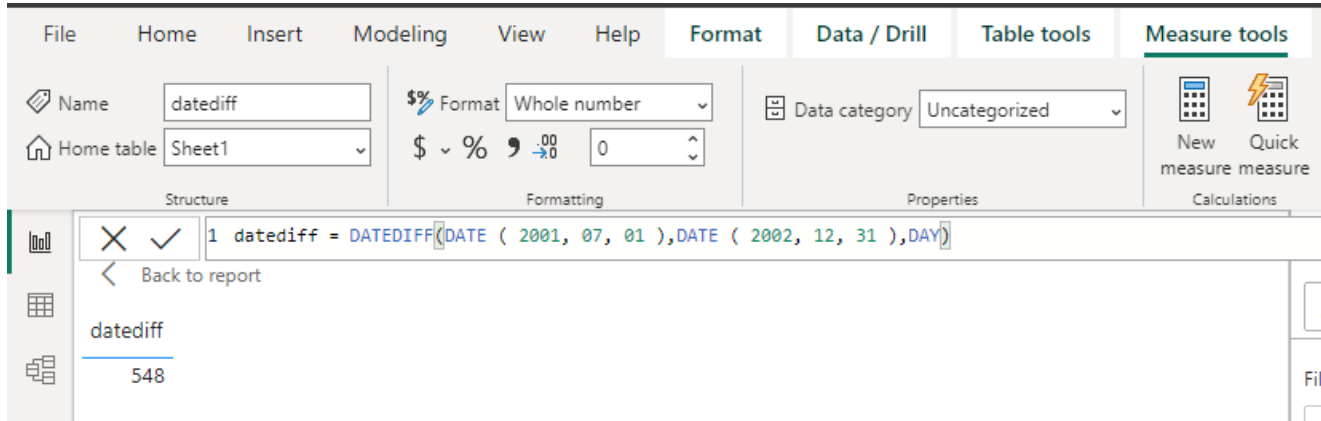
Example:

| datediff = DATEDIFF(DATE (2001, 07, 01),DATE (2002, 12, 31),MONTH)

The screenshot shows the Power BI Desktop interface with the 'Measure tools' tab selected. The 'Name' field is set to 'datediff', the 'Home table' is 'Sheet1', and the 'Format' is 'Whole number'. The 'Data category' is 'Uncategorized'. The measure formula bar displays: `1 datediff = DATEDIFF(DATE ( 2001, 07, 01 ),DATE ( 2002, 12, 31 ),MONTH)`. The 'Structure' pane on the left shows a table with one row and one column labeled 'datediff' with the value '17'. The 'Calculations' pane on the right shows 'New measure' and 'Quick measure' buttons.

Example:

```
datediff = DATEDIFF(DATE (2001, 07, 01),DATE (2002, 12, 31),DAY)
```



DAX Day

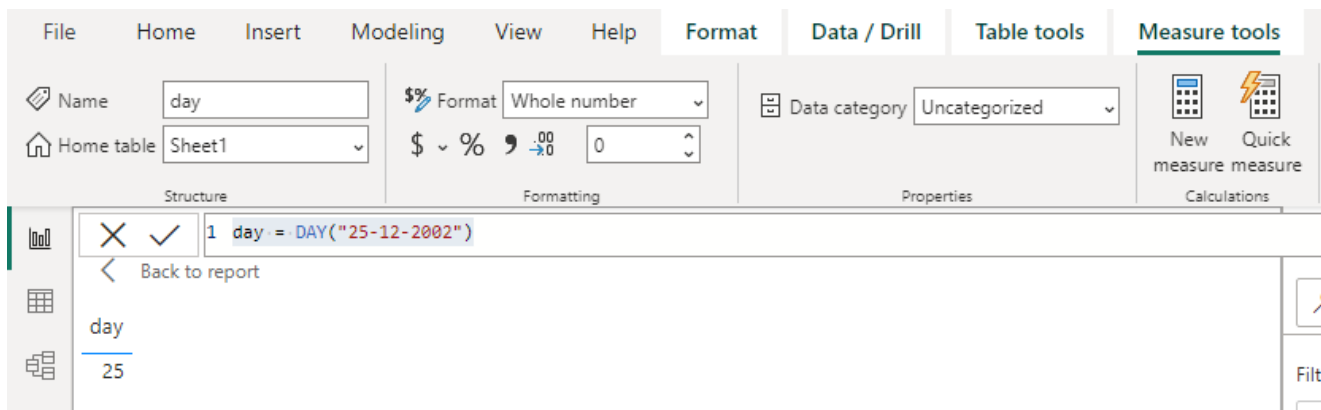
A number between 1 and 31 represents the day of the month. The date of the day you're looking for is an argument for the DAY function. Dates can be passed to the method by inputting them in a DateTime format, using another date function, or using an expression that yields a date. Additionally, you can type a date in one of the acceptable date string formats.

Syntax:

```
DAY(<date>)
```

Example:

```
day = DAY("25-12-2002")
```



DAX Edate

a date that is the specified number of months before or after the start date is returned. To determine maturity dates or due dates that occur on the same day of the month as the date of issue, use the EDATE function.

Syntax:

| EDATE(<start_date>, <months>)

Example:

| edate = EDATE("28-11-1920",4)

The screenshot shows the Microsoft Power BI Desktop interface. The ribbon at the top includes File, Home, Insert, Modeling, View, Help, Format, Data / Drill, Table tools, and Measure tools. The 'Measure tools' tab is active, showing options for 'New measure' and 'Quick measure'. The 'Name' field is set to 'edate', and the 'Home table' is 'Sheet1'. The 'Format' dropdown is set to '*3/14/2001 1:30:55...'. The 'Data category' is 'Uncategorized'. The 'Structure' pane on the left shows a table with one row and one column, with the value '3/28/1921 12:00:00 AM'. The 'Measure tools' pane on the right shows the formula '1 edate = EDATE("28-11-1920",4)' and a 'Back to report' button.

DAX Eomonth

Returns the date of the final day of the month, before or after a specified number of months, in DateTime format. To get maturity dates or due dates that fall on the last day of the month, use the EOMONTH function.

Syntax:

| EOMONTH(<start_date>, <months>)

Example:

| eomonth = EOMONTH("28-11-1920",3)

The screenshot shows the Microsoft Power BI Desktop interface. The ribbon at the top includes File, Home, Insert, Modeling, View, Help, Format, Data / Drill, Table tools, and Measure tools. The 'Measure tools' tab is active, showing options for 'New measure' and 'Quick measure'. The 'Name' field is set to 'eomonth', and the 'Home table' is 'Sheet1'. The 'Format' dropdown is set to '*3/14/2001 1:30:55...'. The 'Data category' is 'Uncategorized'. The 'Structure' pane on the left shows a table with one row and one column, with the value '2/28/1921 12:00:00 AM'. The 'Measure tools' pane on the right shows the formula '1 eomonth = EOMONTH("28-11-1920",3)' and a 'Back to report' button.

DAX Month

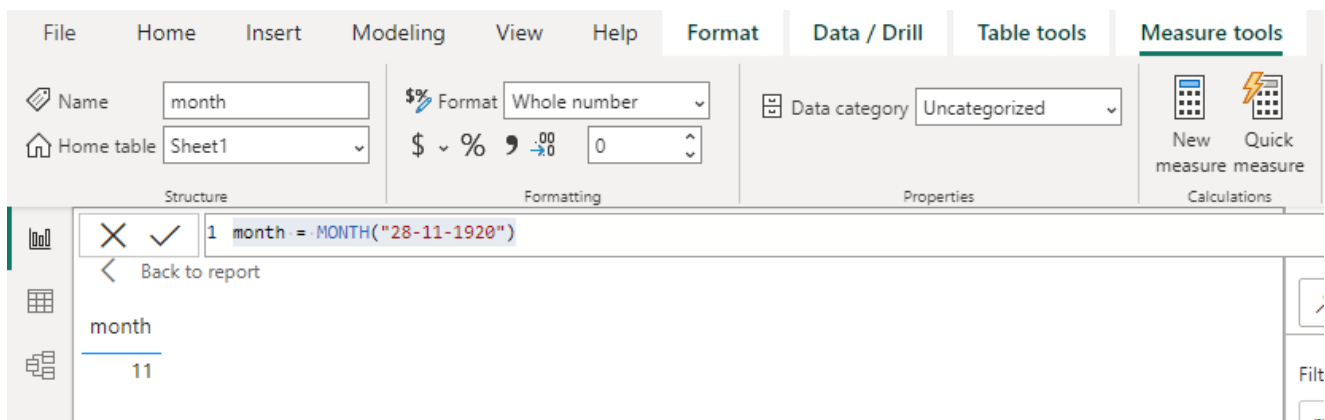
The month is returned as a number ranging from 1 (January) to 12 (December). DAX works with dates in a datetime format as opposed to Microsoft Excel, which maintains dates as serial numbers. You can type a date in a supported datetime format, refer to a column that includes dates, or use an expression that returns a date to enter the date that will be used as the parameter to the MONTH function.

Syntax:

| MONTH(<datetime>)

Example:

| month = MONTH("28-11-1920")



DAX Now

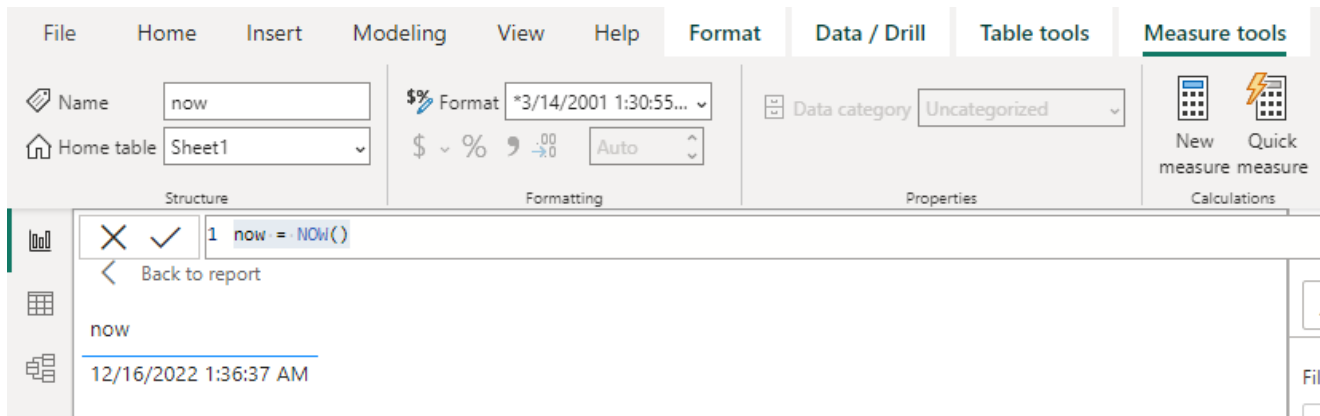
A datetime format string that contains the current date and time. When you need to show the current date and time on a worksheet or compute a number based on the current date and time and have that value updated every time you view the worksheet, the NOW function comes in handy.

Syntax:

| NOW()

Example:

| now = NOW()



DAX Quarter

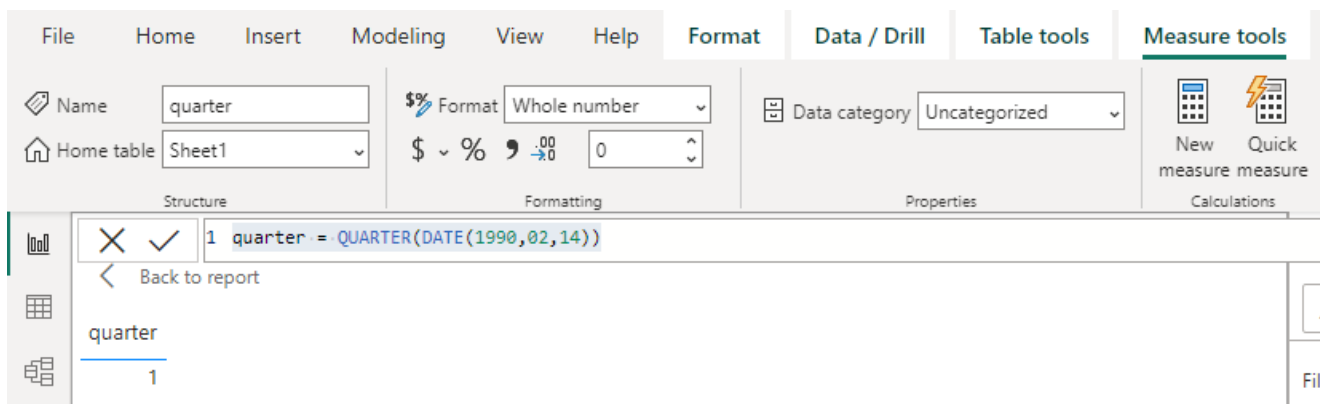
Gives back the quarter as a number between 1 (January through March) and 4. (October – December). The output value is also blank if the input value is blank.

Syntax:

QUARTER(<date>)

Example:

quarter = QUARTER(DATE(1990,02,14))



DAX Today

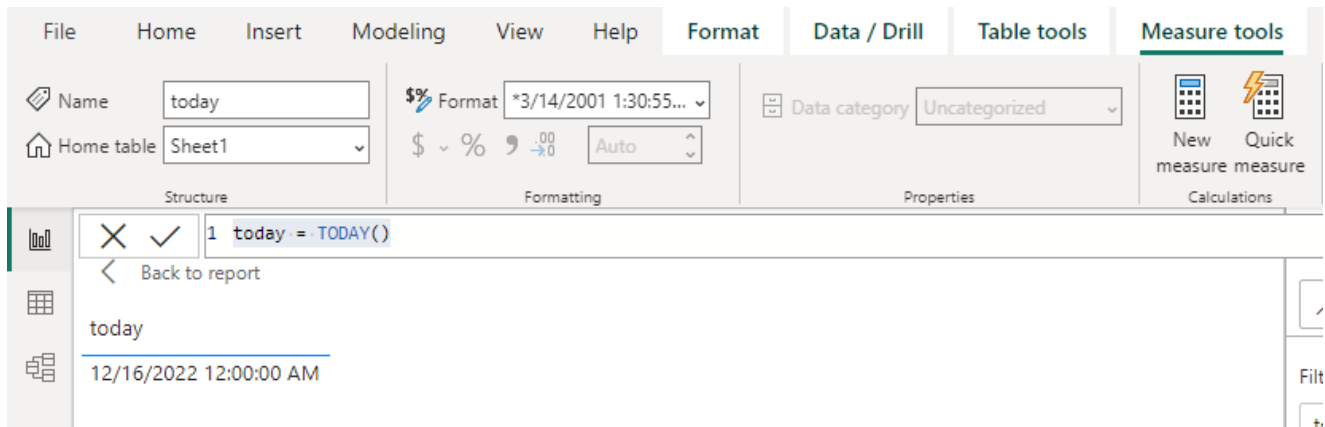
Gives the current date back. The TODAY function comes in handy when you want the current date to be shown on a worksheet regardless of when the workbook is opened. It can be used to compute intervals as well.

Syntax:

TODAY()

Example:

| today = TODAY()



DAX Utcnow

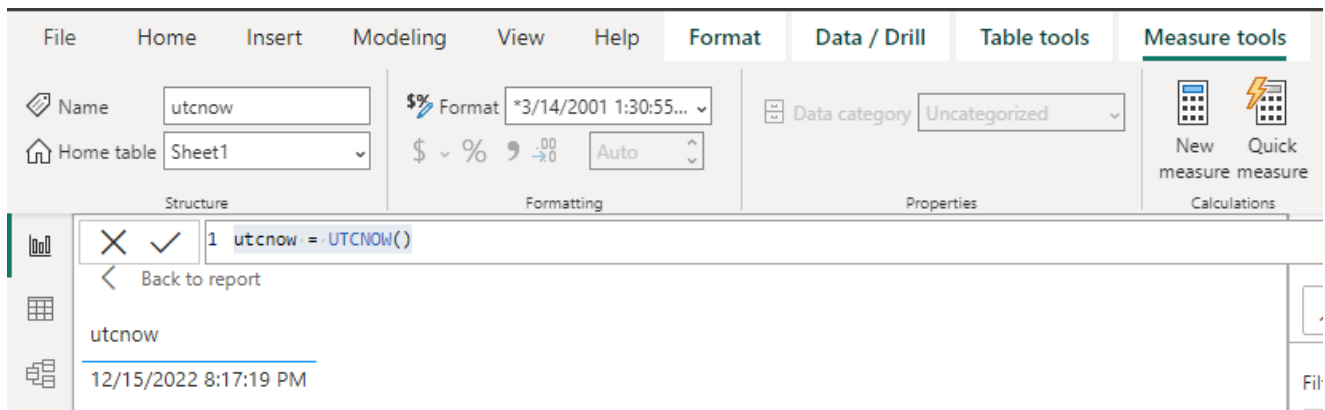
Gives the current date and time in UTC. The UTCNOW function's output only varies when the formula is updated. It isn't always being updated.

Syntax:

| UTCNOW()

Example:

| utcnow = UTCNOW()



DAX UtcToday

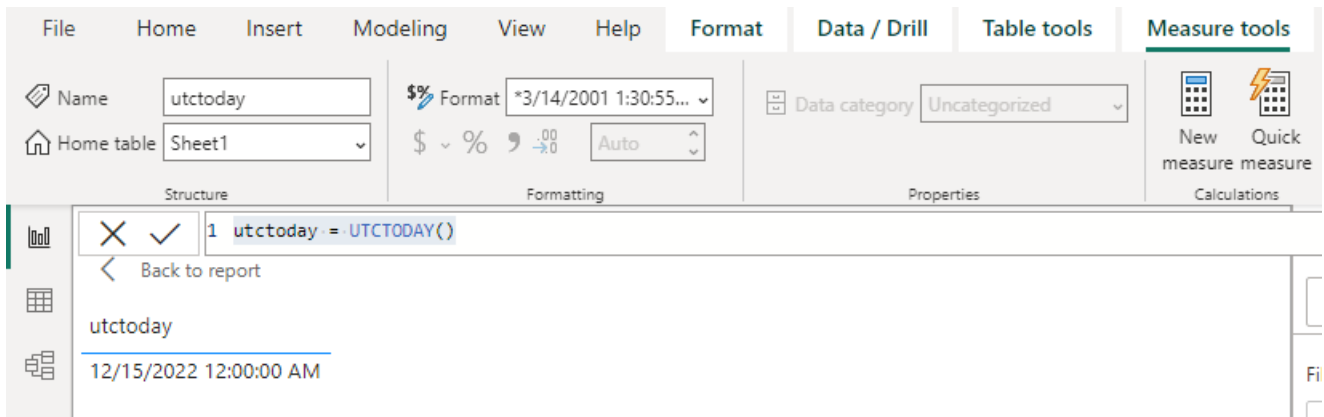
Gives the current date in UTC.

Syntax:

| UTCTODAY()

Example:

| utctoday = UTCTODAY()



DAX Weekday

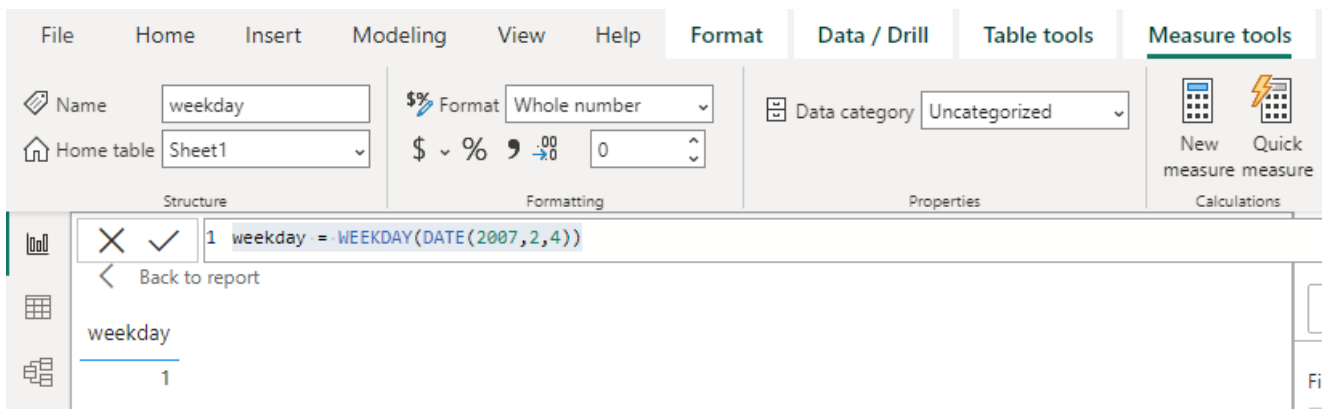
Gives a number between 1 and 7 indicating the date's weekday. The day defaults to being from 1 (Sunday) to 7 (Saturday).

Syntax:

| WEEKDAY(<date>, <return_type>)

Example:

| weekday = WEEKDAY(DATE(2007,2,4))



DAX Weeknum

According to the return type value, returns the week number for the specified date. The week number shows the week's numerical position within a year. For this purpose, two systems are employed:

- **System 1** – Week 1 is the year's first week and corresponds to the week beginning January 1.

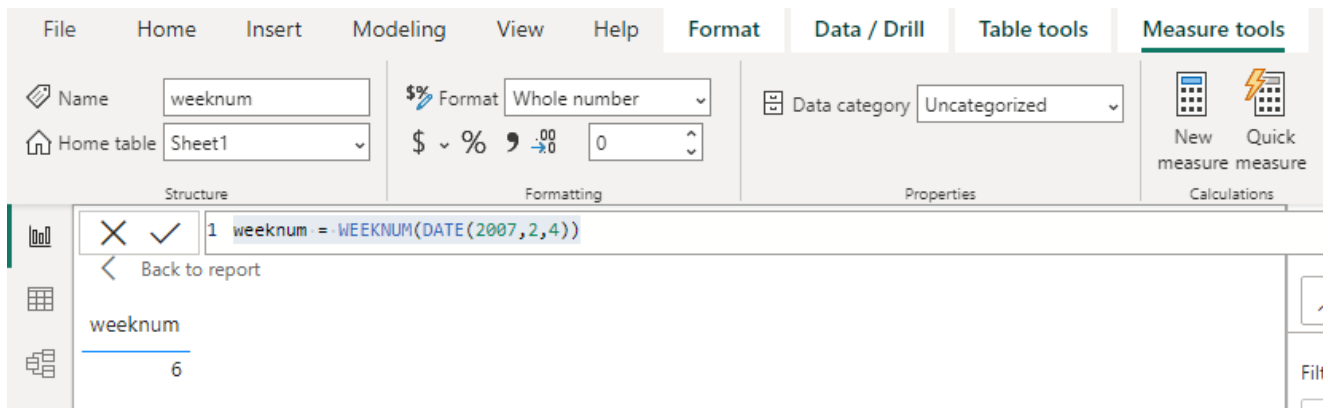
- **System 2** – The week that includes the first Thursday of the year is week 1 and is the first week of the year. The mechanism used in this system is described in ISO 8601, also referred to as the European week numbering scheme.

Syntax:

| WEEKNUM(<date>[, <return_type>])

Example:

| weeknum = WEEKNUM(DATE(2007,2,4))



DAX Year

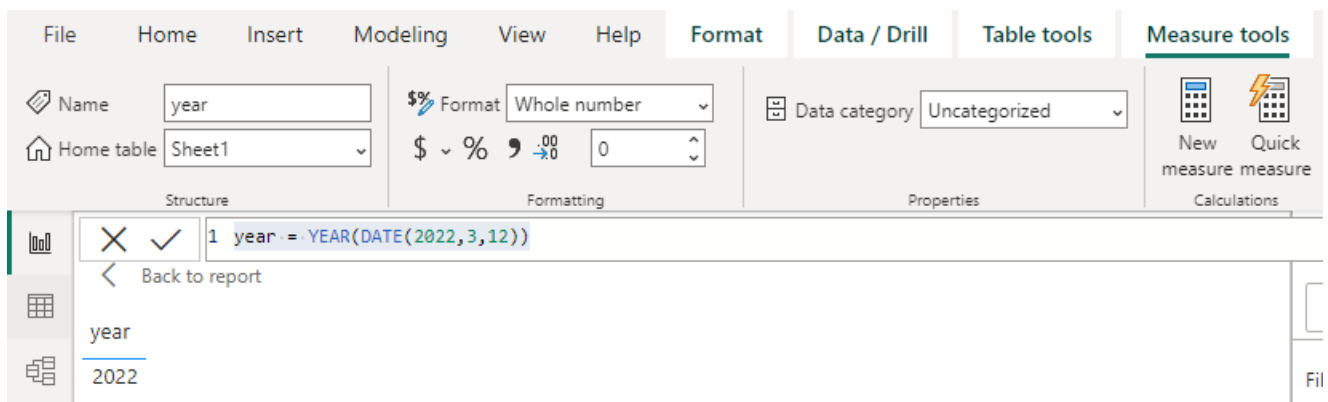
Provides a four-digit integer in the range of 1900 to 9999 as the year of a given date. The date in text or datetime format that includes the desired year.

Syntax:

| YEAR(<date>)

Example:

| year = YEAR(DATE(2022,3,12))



DAX YearFrac

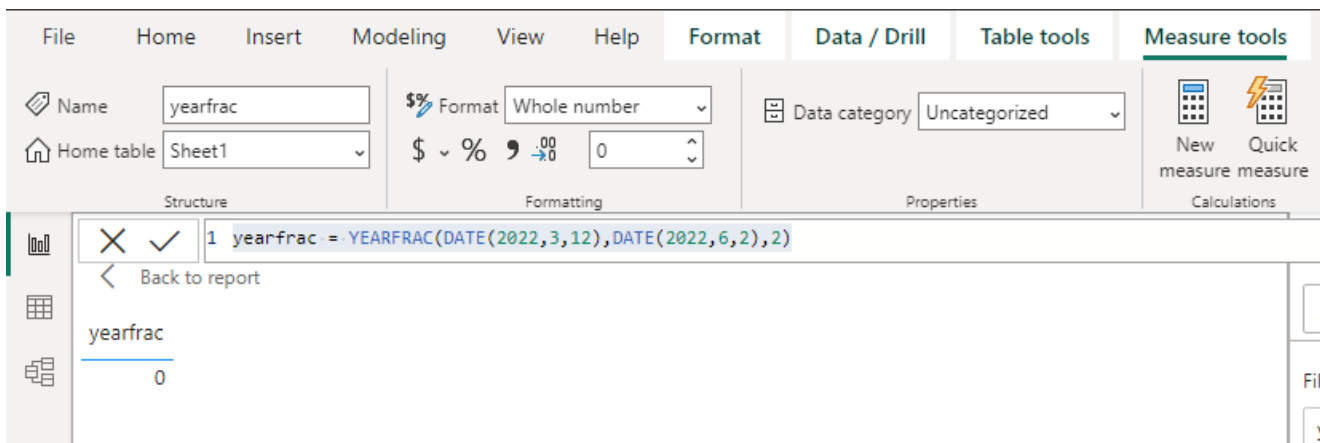
Determines the fraction of the year that the number of full days between two dates corresponds to. To determine the percentage of benefits or responsibilities for an entire year to be assigned to a certain term, use the YEARFRAC spreadsheet function.

Syntax:

| YEARFRAC(<start_date>, <end_date>, <basis>)

Example:

| yearfrac = YEARFRAC(DATE(2022,3,12),DATE(2022,6,2),2)



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